SUDHEER PALAKURLA

AI Engineer | Full-Stack Engineer | AI Solutions Architect

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Summary:

AI Engineer with 8+ years of software engineering experience, specialising in building AI-powered applications, LLM integrations and full-stack systems. Hands-on experience developing RAG pipelines, multi-model orchestration, real-time voice AI agents and API-driven applications using Python, FastAPI, PostgreSQL and modern JavaScript frameworks. Experienced in taking applications from initial requirements and architecture through development, cloud deployment and production support. Combines practical AI engineering with strong frontend, backend and infrastructure knowledge, with a focus on building reliable, maintainable solutions that solve real business problems.

Technical & Leadership Expertise:

- **AI & LLM Engineering:** RAG architecture (PostgreSQL, pgvector, HNSW), prompt engineering, multi-model LLM routing and automatic failover, agentic workflows, MCP, real-time voice AI (OpenAI Realtime API, Gemini Live), guardrails, tokenization/context management and LLM evaluation.
- **Solution Architecture & FDE:** Customer discovery, requirements workshops, workflow analysis, end-to-end solution design, technical specifications, enterprise integration, stakeholder communication, PoC-to-production delivery and customer-environment deployments.
- **Distributed Systems & Messaging:** Microservices, event-driven components, Kafka, RabbitMQ, Redis, Web Sockets, REST APIs, service integration, high-availability design and resilient application communication.
- **Frontend Engineering:** React.js, Next.js, Angular, TypeScript, JavaScript (ES7+), Redux, React Hooks, Vite, HTML5, CSS3, jQuery, Bootstrap, Material UI, WordPress and WooCommerce.
- **Backend & Data:** Node.js, Express.js, Python/FastAPI, PHP, REST API design, PostgreSQL, MySQL, MongoDB, SQL and Alembic schema migrations.
- **Cloud, DevOps & Infrastructure:** AWS, Oracle Cloud Infrastructure (OCI), Oracle Kubernetes Engine (OKE), Kubernetes, Docker, Terraform, Cloudflare Workers/Pages/D1/R2/KV/Workers AI, Jenkins and GitLab CI/CD.
- **Quality, AI Operations & Delivery:** LLM evaluation, tracing, monitoring and cost optimisation; SonarQube, Postman, Jest, React Testing Library, Vitest, Playwright, pytest, BrowserStack, TDD/BDD, Agile/Scrum, Jira and Confluence.

Selected Career Impact:

- Delivered or contributed to 95+ client web and e-commerce builds across UK and international markets.
- Owned technical delivery across customer engagements, from discovery and requirements workshops through solution architecture, enterprise integration and production release.
- Designed and shipped production RAG systems using PostgreSQL, pgvector and HNSW indexing to support AI-assisted search and in-app guidance.
- Implemented multi-model LLM routing, provider fallback, evaluation, tracing, monitoring and cost-optimisation practices for production AI workflows.
- Led legacy-to-React / TypeScript / NextJs / ThreeJs migrations that improved application performance and maintainability by 30–40%; test automation initiatives reduced production defects by approximately 40%.
- Built real-time voice AI using the OpenAI Realtime API with automatic fallback to Gemini Live to maintain conversational continuity during provider outages.

Educational Qualification:

- **Master of Computer Science (Data Science & AI) – Bournemouth University, UK | Jan 2020 – Jan 2022**
- **Master of Business Administration (HR & Finance) – Osmania University, India.**
- **Bachelor of Technology (Computer Science Engineering) – Kakatiya University, India.**

Professional Experience:

Web Designer - Brittania Consultancy Services Limited, UK | April 2024 - Current

- Work across multiple client projects, taking responsibility for technical delivery from initial discussions and requirements gathering through solution design, development, testing and production deployment.
- Design and develop full-stack web applications using React.js, Next.js, TypeScript and Node.js, building responsive interfaces and integrating them with backend services and REST APIs.
- Develop backend applications and services using Python, FastAPI and Node.js, including API development, database integration, authentication and business logic implementation.
- Designed and developed a RAG-based AI assistant for a case-management ERP platform, using PostgreSQL, pgvector and HNSW indexing to retrieve relevant information and provide context-aware responses.
- Implemented multi-model LLM integration with intelligent provider routing and automatic fallback mechanisms, including local model fallback to improve service availability and reduce dependency on individual AI providers.
- Integrated AI capabilities into business applications, working with prompt engineering, context management, API orchestration and response handling to support practical AI-assisted workflows.
- Developed a real-time voice AI assistant using the OpenAI Realtime API, with Gemini Live as an alternative provider to maintain conversational functionality during service interruptions.
- Designed and developed a serverless client-onboarding platform using Cloudflare Pages, Workers and Hono, with Cloudflare D1 for relational data, KV for caching and R2 for object storage.
- Integrated Cloudflare Workers AI into application workflows to provide AI-assisted functionality while evaluating model capabilities, performance and operating costs.
- Manage application deployments and cloud infrastructure across Oracle Cloud Infrastructure (OCI), AWS and Cloudflare, working with containerisation, environment configuration, service integration and infrastructure automation.
- Implement software quality and testing practices using SonarQube, Jest, Vitest and Playwright, identifying defects, improving application reliability and maintaining release quality.
- Work directly with clients and stakeholders to understand business requirements, discuss technical approaches, prepare implementation plans and resolve application and integration challenges.
- Use Git-based version control, CI/CD workflows, Jira, Confluence and collaboration tools to manage development activities, coordinate releases and maintain technical documentation.

Senior Frontend Developer - Sharp Gaming Limited, UK | Oct 2022 – June 2023

- Led the migration of legacy JavaScript applications to React and TypeScript, improving maintainability, scalability and code organisation.
- Developed reusable React components and frontend modules to support consistent application development.
- Integrated REST APIs and implemented application state management using Redux and React Hooks.
- Collaborated with product, backend, QA and DevOps teams to translate requirements into technical solutions and deliver application enhancements.

- Investigated and resolved complex frontend defects, integration issues and cross-browser compatibility problems.
- Conducted code reviews, supported developer knowledge sharing and applied SonarQube and testing practices to maintain code quality.
- Contributed to technical planning, task estimation and frontend performance optimisation, helping improve delivery efficiency and application reliability.

Senior Developer - Crazy Techsol Pvt Ltd, India | May 2017 – Jan 2020

- Led delivery across multiple concurrent client projects, coordinating requirements, frontend development, API integration, testing and stakeholder updates.
- Spearheaded migration of applications from PHP/jQuery to React, TypeScript and Redux, reducing technical debt and improving scalability.
- Built full-stack e-commerce platforms (frontend and backend) for 8+ client stores spanning clothing and bio/wellness products, using PHP for some projects and Node.js for others.
- Established reusable component libraries, code-review practices, version-control standards and CI processes adopted across the team.
- Authored best-practice manuals and technical documentation that streamlined workflows for cross-functional teams.
- Mentored junior developers and led client meetings to align technical delivery with business and commercial objectives.

Web Designer - Crazy Designers, India | Dec 2014 – Apr 2017

- Designed and delivered 40+ responsive websites and e-commerce solutions end-to-end using WordPress, WooCommerce, Magento, HTML5/CSS3, JavaScript and Bootstrap.
- Created UI mockups and visual design assets in Adobe Photoshop, translating them into responsive WordPress and static HTML/CSS website builds.
- Worked directly with clients to gather requirements, build prototypes and translate business needs into practical digital solutions.
- Built payment gateway integrations and custom CMS themes, improving performance, usability and cross-browser compatibility.

Independent Project – AI-Driven SaaS Platform | Sole Engineer

AI Engineering, Solution Architecture & Full-Stack Development

Technologies: Python, FastAPI, PostgreSQL, pgvector, HNSW, RAG, LLM Integration, Multi-Model Routing, OpenAI Realtime API, Gemini Live, Alembic, Oracle Cloud Infrastructure (OCI).

- Architected and developed an AI-driven SaaS platform end to end, taking ownership of solution design, technical architecture, backend development, AI integrations and cloud deployment.
- **Real-Time Voice AI Agent:** Integrated the OpenAI Realtime API with Gemini Live fallback to enable real-time conversational voice interactions.
- **RAG-Based AI Assistant:** Designed and implemented a retrieval-augmented generation pipeline using PostgreSQL, pgvector and HNSW indexing to provide context-aware information retrieval and AI-powered assistance.

- **Multi-Model LLM Orchestration:** Implemented intelligent model routing across hosted LLM providers, with fallback to a local model to improve service resilience and reduce dependency on individual providers.
- Developed backend services and database workflows, managing schema changes and version-controlled migrations using Alembic.
- Managed cloud deployment and application configuration on Oracle Cloud Infrastructure (OCI), covering environment setup, integration and application delivery.

Selected Projects:

Betfred.com — Enterprise Betting and Gaming Platform

Technologies: Kafka, RabbitMQ, Redis, Instana, OKE/Kubernetes, Terraform, WebSockets.

- Contributed to a large-scale online betting and gaming platform involving distributed services and real-time application functionality.
- Worked with Kafka and RabbitMQ for messaging, Redis for caching, and WebSockets for real-time communication.
- Supported Kubernetes/OKE infrastructure, Terraform-based provisioning and Instana application performance monitoring.
- Contributed to payment integrations, microservices and reliable application delivery.

GetMyHotels — AI-Powered Hotel Booking Platform

Technologies: Kafka, Redis, Terraform, WebSockets, Payment Integrations.

- ***Developed functionality for a hotel booking platform supporting customer-facing search and booking workflows.***
- ***Worked with Kafka-driven messaging, Redis caching and WebSockets for application communication.***
- ***Integrated payment functionality and supporting backend services.***
- ***Supported infrastructure provisioning and deployment using Terraform.***

3 Bolt Court — Legal Services Platform

Technologies: Redis, Microservices, Terraform, Payment Integrations.

- Delivered a digital platform for a professional legal services organisation.
- Developed application functionality using service-oriented architecture and Redis caching.
- Integrated payment-related functionality and supporting backend components.
- Supported infrastructure provisioning and application deployment using Terraform.

Other selected projects: Brittania Consultancy, SK Security Services, i20Fever, Dr Foot App, Crazy Techsol, KR Energy Consultants, KSRG Energy, Future Cross Academy and Sanguine Bio.